Azure Interview Questions & Answers for Beginners

Beginner-level Azure interview preparation for Cloud Engineer roles

1. What is Microsoft Azure?

Answer: Microsoft Azure is a **cloud computing platform** provided by Microsoft. It provides services such as virtual machines, storage, databases, networking, security, monitoring, and serverless computing.

Simple example: Instead of buying a physical server, we can create a **Virtual Machine in Azure** and use it through the internet.

2. What is Cloud Computing?

Answer: Cloud computing means using computing resources such as **servers, storage, databases, and networking over the internet** instead of managing physical hardware ourselves.

For example, Azure provides servers that we can create and use whenever required.

3. What is an Azure Resource?

Answer: A resource is anything that we create and use in Azure.

Examples:
- Virtual Machine
- Storage Account
- Virtual Network
- SQL Database
- Load Balancer
- Azure Function

4. What is a Resource Group?

Answer: A Resource Group is a **logical container used to manage Azure resources**.

For example, if we create an application containing a VM, Storage Account, Database, and Load Balancer, we can put all of them inside one Resource Group.

Interview point: Resources in a Resource Group can be managed together.

5. What is an Azure Region?

Answer: An Azure Region is a **geographical location where Azure has datacenters**.

Examples:
- Central India
- South India
- East US
- West Europe

We select a region based on requirements such as latency, availability, compliance, and cost.

6. What is an Availability Zone?

Answer: An Availability Zone is a **physically separate datacenter within an Azure region**.

If one zone has a failure, applications running in another zone can continue operating.

Example:
Azure Region
■■■ Zone 1
■■■ Zone 2
■■■ Zone 3

7. What is an Azure Virtual Machine?

Answer: An Azure VM is a **virtual server running in the Azure cloud**.

We can install operating systems such as:
• Linux
• Windows Server

We can use a VM to host applications, websites, databases, or other workloads.

8. What is an Azure Storage Account?

Answer: A Storage Account provides **cloud storage in Azure**.

It can contain services such as:
• Blob Storage
• Azure Files
• Queue Storage
• Table Storage

Example: We can use Blob Storage to store images, videos, backups, and documents.

9. What is Azure Blob Storage?

Answer: Azure Blob Storage is an **object storage service** used to store large amounts of unstructured data.

Examples:
• Images
• Videos
• Documents
• Log files
• Backups

10. What is Azure Virtual Network?

Answer: Azure Virtual Network, or **VNet**, is a private network in Azure.

It allows Azure resources such as VMs to communicate with each other securely.

Example:
VNet
■■■ Subnet 1 → Web VM
■■■ Subnet 2 → Database VM

11. What is a Subnet?

Answer: A subnet is a **smaller network inside a VNet**.

Example:
VNet: 10.0.0.0/16
Subnet-Web: 10.0.1.0/24
Subnet-App: 10.0.2.0/24
Subnet-DB: 10.0.3.0/24

We can use different subnets to organize and isolate different workloads.

12. What is an NSG in Azure?

Answer: NSG stands for **Network Security Group**.

It controls network traffic going **into and out of Azure resources**.

For example, we can create a rule:
• Allow TCP 80
• Allow TCP 443
• Deny TCP 22

NSGs are commonly associated with **subnets or network interfaces**.

13. What is Azure Load Balancer?

Answer: Azure Load Balancer distributes incoming network traffic across multiple backend resources.

Example:
Users → Azure Load Balancer → VM1 / VM2

If many users access an application, the Load Balancer distributes traffic between the VMs.

14. What is Azure Application Gateway?

Answer: Application Gateway is a **Layer 7 web traffic load balancer**.

It can provide features such as:
• HTTP/HTTPS routing
• URL-based routing
• SSL/TLS termination
• Web Application Firewall (WAF)

Simple difference:
Azure Load Balancer → Layer 4
Application Gateway → Layer 7

15. What is Azure DNS?

Answer: Azure DNS is a DNS hosting service provided by Azure.

It translates domain names into IP addresses.

Example:
www.example.com → DNS lookup → IP Address

16. What is Azure IAM?

Answer: Azure uses **Microsoft Entra ID** for identity management and **Azure RBAC** for authorization to Azure resources.

RBAC stands for **Role-Based Access Control**.

It determines **who can do what on which resource**.

Example:
User → Developer → Can read resources → Cannot delete resources

17. What is Microsoft Entra ID?

Answer: Microsoft Entra ID is Microsoft's **cloud-based identity and access management service**.

It manages:
- Users
- Groups
- Applications
- Authentication
- Access

It was previously called **Azure Active Directory (Azure AD)**.

18. What is Azure RBAC?

Answer: Azure RBAC controls what actions a user, group, service principal, or managed identity can perform on Azure resources.

It is used to implement **least-privilege access**.

19. Difference between Owner, Contributor and Reader?

Answer:

Owner: Can manage resources and assign access.

Contributor: Can create, modify and delete resources but cannot normally assign Azure RBAC permissions.

Reader: Can view resources but cannot modify them.

20. What is Azure Monitor?

Answer: Azure Monitor is a monitoring service used to collect and analyze information about Azure resources and applications.

It can help us monitor:
• CPU usage
• Memory-related metrics where available
• Network activity
• Application performance
• Logs
• Alerts

21. What is an Azure Alert?

Answer: An Azure Alert automatically notifies us when a particular condition occurs.

Example:
VM CPU > 80% → Azure Alert → Notification

This helps administrators detect problems quickly.

22. What is Azure Backup?

Answer: Azure Backup is a service used to **backup Azure resources and data**.

For example, we can use Azure Backup to protect Azure VMs.

23. What is a Recovery Services Vault?

Answer: A Recovery Services vault is a storage and management location used by **Azure Backup and Azure Site Recovery**.

It can be used to manage:
• VM backups
• Backup policies
• Recovery points
• Disaster recovery operations

24. What is Azure Site Recovery?

Answer: Azure Site Recovery is a **disaster recovery service**.

It helps replicate workloads and recover them when a primary site or region becomes unavailable.

Example:
Primary Region → Replication → Secondary Region

25. What is Azure SQL Database?

Answer: Azure SQL Database is a **fully managed relational database service** based on Microsoft SQL Server.

Microsoft manages many infrastructure tasks such as:
• Hardware
• Operating system
• Patching
• Backups
• High availability

26. What is Azure Functions?

Answer: Azure Functions is a **serverless compute service**.

We can execute code without managing a virtual machine.

Functions can be triggered by events such as:
• HTTP request
• Timer
• Queue message
• Blob event

Example:
File uploaded to Blob → Azure Function → Process the file

27. What is Azure Container Registry?

Answer: Azure Container Registry, or **ACR**, is a private registry for storing container images.

Example:
Docker Build → Docker Image → Azure Container Registry → AKS

28. What is AKS?

Answer: AKS stands for **Azure Kubernetes Service**.

It is a managed Kubernetes service provided by Azure.

It is used to deploy and manage containerized applications.

29. What is Azure App Service?

Answer: Azure App Service is a **managed platform for hosting web applications and APIs**.

We don't need to manually manage the underlying servers.

It supports technologies such as:
• .NET
• Java
• Node.js
• Python
• PHP

30. What is the difference between VM and App Service?

Answer:

Virtual Machine: We have more control over the operating system and infrastructure, but we are responsible for more management.

App Service: Azure manages much of the underlying infrastructure, so we can focus more on the application.

31. A VM is not accessible. What will you check?

Answer: I would check step by step:

1. Is the VM running?
2. Is the VM's IP address correct?
3. Is the required port open?
4. Check NSG rules.
5. Check whether the application/service is running.
6. Check the operating system firewall.
7. Check routing and subnet configuration.
8. Check Azure Monitor/logs if required.

32. How would you secure an Azure VM?

Answer: I would:

• Use NSGs to control network traffic.
• Avoid exposing unnecessary ports to the internet.
• Use SSH keys for Linux where appropriate.
• Use Microsoft Entra ID-based access where appropriate.
• Keep the OS patched.
• Use Azure Bastion or another controlled access method where appropriate.
• Monitor the VM using Azure Monitor.
• Follow least-privilege access using Azure RBAC.

33. Your application is receiving a lot of traffic. What would you do?

Answer: I would consider **scaling the application**.

For example:
Users → Load Balancer → VM1 / VM2 / VM3

I could use **VM Scale Sets** or a managed application platform depending on the workload.

34. How would you store images uploaded by users?

Answer: I would use **Azure Blob Storage** because it is designed for storing object data such as images, videos, and documents.

35. How would you backup an Azure VM?

Answer: I would use **Azure Backup** and configure a backup policy in a **Recovery Services vault**.

36. What is the difference between Azure and AWS?

Answer: Both are cloud platforms that provide services such as compute, storage, networking, databases, security, and monitoring.

Some common equivalents are:
AWS EC2 → Azure Virtual Machines
S3 → Blob Storage
VPC → VNet
IAM → Entra ID + Azure RBAC
ELB → Azure Load Balancer
RDS → Azure SQL Database / Azure Database services
Lambda → Azure Functions
EKS → AKS
CloudWatch → Azure Monitor
ECR → Azure Container Registry

Important Scenario-Based Questions

31. A VM is not accessible. What will you check?

Check whether the VM is running, whether the IP address is correct, whether the required port is open, NSG rules, whether the application/service is running, the operating system firewall, routing and subnet configuration, and Azure Monitor/logs if required.

32. How would you secure an Azure VM?

Use NSGs to control network traffic, avoid exposing unnecessary ports, use SSH keys for Linux where appropriate, use Microsoft Entra ID-based access where appropriate, keep the OS patched, use Azure Bastion or another controlled access method where appropriate, monitor the VM, and follow least-privilege access using Azure RBAC.

33. Your application is receiving a lot of traffic. What would you do?

Consider scaling the application. For example, a Load Balancer can distribute traffic across multiple VMs. VM Scale Sets or managed application platforms can also be considered depending on the workload.

34. How would you store images uploaded by users?

I would use Azure Blob Storage because it is designed for storing object data such as images, videos, and documents.

35. How would you backup an Azure VM?

I would use Azure Backup and configure a backup policy in a Recovery Services vault.

36. What is the difference between Azure and AWS?

Both are cloud platforms providing compute, storage, networking, databases, security, and monitoring. Common equivalents include AWS EC2 → Azure Virtual Machines, S3 → Blob Storage, VPC → VNet, Lambda → Azure Functions, EKS → AKS, CloudWatch → Azure Monitor, and ECR → Azure Container Registry.

Azure vs AWS Quick Comparison

AWS	Azure
EC2	Virtual Machines
S3	Blob Storage
VPC	VNet
IAM	Entra ID + Azure RBAC
ELB	Azure Load Balancer
RDS	Azure SQL Database / Azure Database services
Lambda	Azure Functions
EKS	AKS
CloudWatch	Azure Monitor
ECR	Azure Container Registry

Questions You Should Definitely Prepare

1. What is Azure?
2. What is Resource Group?
3. What is Region?
4. What is Availability Zone?
5. What is Azure VM?
6. What is VNet?
7. What is Subnet?
8. What is NSG?
9. What is Load Balancer?
10. What is Application Gateway?
11. What is Blob Storage?
12. What is Azure SQL Database?
13. What is Entra ID?
14. What is Azure RBAC?
15. Difference between Owner, Contributor and Reader
16. What is Azure Monitor?
17. What is Azure Backup?
18. What is Recovery Services Vault?

19. What is Azure Functions?
20. What is AKS?
21. What is ACR?
22. What is App Service?
23. Azure VM troubleshooting
24. How to secure a VM?
25. Azure vs AWS

Interview Tip: Don't just memorize definitions. For each Azure service, be ready to explain **what it is + why we use it + one practical example**. This makes your answer sound more like a real Cloud Engineer interview response.